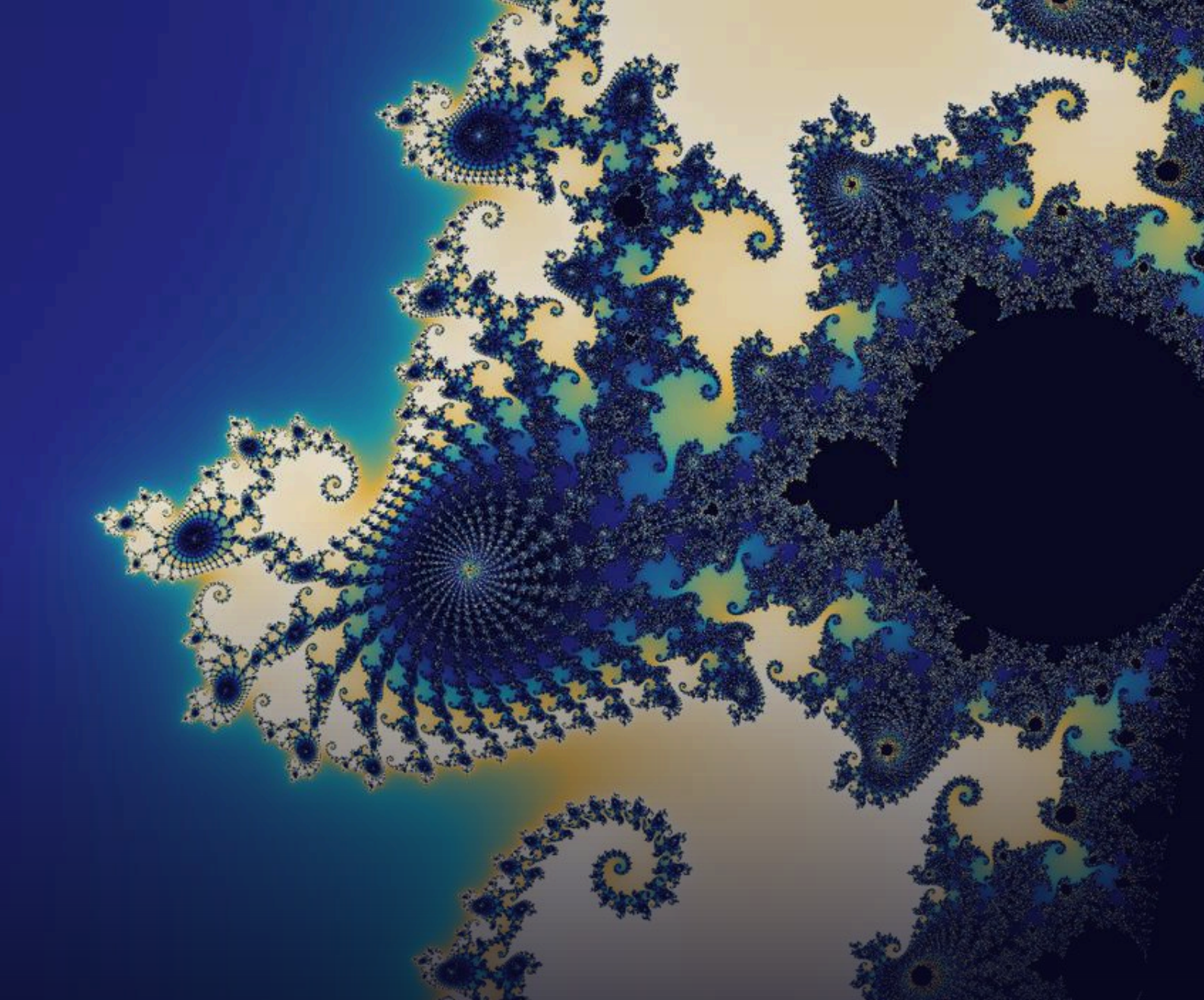


KINGSCLIFF HIGH SCHOOL · MATHEMATICS

MANDELBROT SERIES

Bridge to Year 9 Advanced

Seven lessons of revision and consolidation of the Year 8 algebra, indices, Pythagoras and measurement skills that Year 9 Advanced is built on.

1 Algebra basics

2 Expanding brackets

3 Factorising

4 Indices

5 Square roots & Pythagoras' theorem

6 Equations & formulas

7 Area & volume

NAME _____ CLASS _____ TEACHER _____

How this booklet works

One lesson = three pages = three rounds

Each lesson starts with **key ideas** and **worked examples**, then three rounds of questions that get harder as you go.

- **B** **Bronze – support.** Quick drill questions. Everyone starts here; most should be done without help.
- **S** **Silver – core.** Multi-step drill and worded problems. This is the standard for Year 9 Advanced.
- **G** **Gold – extension.** Harder problems and a preview of Year 9 ideas. Try them once Silver is done.

Mixed questions

Questions with a chip like ↻ L2 come from an **earlier lesson**; chips like ↻ Percentages revisit **other Year 8 topics**. Mixing topics feels harder, but it makes what you learn last much longer.

Symbols

★ marks a Silver question that goes beyond Stage 4 core (it is in your Year 8 textbook as starred or Stage 5 work). All Gold questions are extension.

No calculators. Leave answers exact (e.g. $\sqrt{74}$ or 90π) unless a question asks you to round. Write your working in the spaces provided.

My progress

LESSON	SYLLABUS	B	S	G	CONFIDENCE (CIRCLE)
1 Algebra basics	MA4-ALG-C-01	☐	☐	☐	1 2 3 4
2 Expanding brackets	MA4-ALG-C-01	☐	☐	☐	1 2 3 4
3 Factorising	MA4-ALG-C-01	☐	☐	☐	1 2 3 4
4 Indices	MA4-IND-C-01	☐	☐	☐	1 2 3 4
5 Square roots & Pythagoras' theorem	MA4-IND-C-01 (roots) · MA4-PYT-C-01	☐	☐	☐	1 2 3 4
6 Equations & formulas	MA4-EQU-C-01	☐	☐	☐	1 2 3 4
7 Area & volume	MA4-ARE-C-01 · MA4-VOL-C-01	☐	☐	☐	1 2 3 4

Pre-test / 20 | Post-test / 20

A 50-minute lesson

0–5	5–18	18–40	40–50
Key ideas & examples	Bronze round	Silver round	Gold · check answers

Finished early? Go straight to Gold. Stuck? Re-read the worked examples, then ask.

Getting the most out of it

- Try every question before asking for help – a wrong attempt teaches more than a blank.
- Show your working: in Year 9 Advanced most marks are for method.
- Check answers with your teacher; circle any you got wrong and redo them a few days later.
- Mixed (↻) questions are meant to make you stop and think "which skill is this?"

Why Mandelbrot?

Every banner in this booklet is a zoom into the Mandelbrot set, drawn by repeating one tiny rule, $z \to z^2 + c$, over and over. Simple algebra, repeated carefully, builds something extraordinary – which is exactly what Year 9 Advanced will ask of you.

BRIDGE TO YEAR 9 ADVANCED

Pre-test

Do this before Lesson 1. It shows what you already know and which rounds to focus on.

20 marks · 20 minutes · No calculator

Name _____ Date _____ **Score** / 20

1 Algebra basics

Lesson 1 · 3 marks

a) Simplify $5x + 3y - 2x$

.....[1]

b) Simplify $4a \times 3ab$

.....[1]

c) Evaluate $2p^2 - q$ when $p = -3$ and $q = 4$

.....[1]

2 Expanding brackets

Lesson 2 · 3 marks

a) Expand $x(x - 4)$

.....[1]

b) Expand $-3(2x - 5)$

.....[1]

c) Expand and simplify $4(a + 2) - 2(a - 3)$

.....[1]

3 Factorise fully

Lesson 3 · 3 marks

a) $12m + 18$

.....[1]

b) $6x^2 - 9xy$

.....[1]

c) $-4a - 10$

.....[1]

4 Indices

Lesson 4 · 3 marks

a) Write $5^7 \times 5^2 \div 5^6$ as a single power, then evaluate

.....[1]

b) Evaluate $(2^3)^2 \div 2^4$

.....[1]

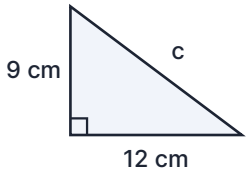
c) Write 180 as a product of prime factors in index form

.....[1]

5 Pythagoras' theorem

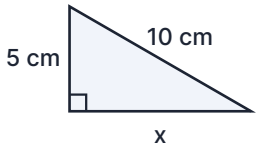
Lesson 5 · 3 marks

a) Find c.



[1]

b) Find x as an exact value.



[1]

c) Is a triangle with sides 5, 11 and 12 right-angled? Show working.

[1]

6 Solve

Lesson 6 · 3 marks

a) $4x - 7 = 13$

[1]

b) $5a + 3 = 2a - 9$

[1]

c) $2x^2 = 72$

[1]

7 Area and volume

Lesson 7 · 2 marks

a) Find the area of a trapezium with parallel sides 5 cm and 9 cm and perpendicular height 4 cm.

[1]

b) Find the volume of a cylinder with radius 3 cm and height 5 cm, in terms of π.

[1]

Where should I start each lesson?

QUESTION	LESSON	MY MARKS	START AT…
1 Algebra basics	Lesson 1	/ 3	Bronze ☐ Silver ☐ Gold ☐
2 Expanding brackets	Lesson 2	/ 3	Bronze ☐ Silver ☐ Gold ☐
3 Factorise fully	Lesson 3	/ 3	Bronze ☐ Silver ☐ Gold ☐
4 Indices	Lesson 4	/ 3	Bronze ☐ Silver ☐ Gold ☐
5 Pythagoras' theorem	Lesson 5	/ 3	Bronze ☐ Silver ☐ Gold ☐
6 Solve	Lesson 6	/ 3	Bronze ☐ Silver ☐ Gold ☐
7 Area and volume	Lesson 7	/ 2	Bronze ☐ Silver ☐ Gold ☐

Full marks on a question? Move quickly through Bronze and spend your time on Silver and Gold. Lost marks? Do every Bronze question first.

LESSON 1 OF 7

Algebra basics

Like terms · multiplying and dividing terms · substitution · simple algebraic fractions

50 min

No calculator

Outcomes: MA4-ALG-C-01 · MAO-WM-01 | Gold: MA5-ALG-C-01 (algebraic fractions with numerical denominators) · MA5-ALG-P-01/P-02 (indices, binomial numerators)

NCM 8 Adv 3.01–3.05 → NCM 9 Adv 1.01–1.04

KEY IDEAS

- **Like terms** have exactly the same pronumerals with the same powers: $3x^2y$ and $-5x^2y$ are like terms; $3x^2$ and $3x$ are not.
- **Add or subtract** only like terms: $5a + 2b + 3a = 8a + 2b$.
- **Multiply** the numbers, then the pronumerals: $4a \times 3ab = 12a^2b$.
- **Divide** by writing a fraction and cancelling: $12xy \div 4x = \frac{12xy}{4x} = 3y$.
- **Substitute** using brackets: if $a = -3$, then $a^2 = (-3)^2 = 9$ but $-a^2 = -9$.

WORKED EXAMPLES

Example 1 Simplify $5x + 3y - 2x + 7y$.

$= 5x - 2x + 3y + 7y$

$= 3x + 10y$

Example 2 Evaluate $2a^2 - 3b$ when $a = -3,\ b = 4$.

$= 2(-3)^2 - 3(4)$

$= 18 - 12 = 6$

Example 3 Simplify $\frac{2x}{3} \times \frac{9}{4x}$.

$= \frac{18x}{12x} = \frac{3}{2}$

B Bronze round

Support · everyone starts here

Simplify.

1 $7a + 3a$

2 $9m - 4m$

3 $5x + 2y + 3x$

4 $6p - p + 2q$

5 $4 \times 3k$

6 $5a \times 2b$

7 $3m \times m$

8 $12y \div 3$

9 $20ab \div 5b$

10 Evaluate $-7 + 12$

Integers

11 $8x - 3 - 2x + 7$

12 $-3 \times 4n$

13 Evaluate $\frac{3}{4} \times 36$

Fractions

14 $2ab + 5ab - ab$

If $a = 4$, $b = -2$ and $c = 0.5$, evaluate:

15 $3a + b$

16 ab

17 $a^2 + b^2$

18 $10c$

19 25% of 80

Percentages

20 $b^2 - a$

S Silver round

Core · aim to finish this round

Simplify.

21 $3x^2 + 5x - x^2 + 2x$

22 $4ab - 3ba + 2a$

23 $-6m \times (-2mn)$

24 $2xy \times 3xy$

25 $-18pq \div 6q$

26 $15a^2b \div 5ab$

27 $\dfrac{8xy}{12y}$

28 Simplify the ratio $18 : 24$

↻ Ratios

29 $5x - (-3x)$

30 $\dfrac{x}{3} \times 6$

31 $\dfrac{2a}{5} \times \dfrac{10}{a}$

32 $\dfrac{3m}{4} \div \dfrac{m}{8}$

33 If $y = 3x - 5$, find y when $x = -2$

↻ Linear

34 Increase $\$60$ by 15%

↻ Percentages

35 Evaluate $-2 \times (-3) \times (-5)$

↻ Integers

If $p = -3$ and $q = 5$, evaluate:

36 $p^2 - 2q$

37 $-p^2$

38 $(p + q)^2$

39 $\dfrac{q - p}{4}$

40 Pizzas cost $\$14$ each, plus a $\$6$ delivery fee.
a) Write an expression for the cost, in dollars, of n pizzas.
b) Find the cost of 3 pizzas.

41 Mia says $3x + 2y = 5xy$. Choose values for x and y to show she is wrong, then explain her mistake.

Gold round

Extension · stretch yourself

Simplify.

42 $\frac{x}{3} + \frac{x}{4}$

43 $\frac{2a}{5} - \frac{a}{3}$

44 $\frac{5m}{6} - \frac{m}{4}$

45 $\frac{6x^2y}{9xy^2}$

46 $\frac{3x}{4} - \frac{x}{6}$

47 $\frac{4ab}{3} \times \frac{9}{2b}$

48 $\frac{2m}{3} \div \frac{4m}{9}$

49 $\frac{x+1}{2} + \frac{x}{3}$

50 A rectangle has length $3x + 2$ and width $x - 1$.

a) Show that its perimeter is $8x + 2$.

b) Find the perimeter when $x = 4$.

c) Why must x be greater than 1?

51 Without a calculator, evaluate $a^2 - 2ab + b^2$ when $a = 7.5$ and $b = 2.5$.

Hint: look for a shortcut.

52 The nth term of a pattern is $3n^2 - n$.

a) Find the 10th term.

b) Is 70 a term of the pattern? Justify your answer.

53 Two numbers a and b have $a + b = 10$ and $ab = 21$. Find the value of $a^2 + b^2$.

LESSON 2 OF 7

Expanding brackets

The distributive law · expanding and simplifying · binomial products (Gold)

50 min

No calculator

Outcomes: MA4-ALG-C-01 · MAO-WM-01 | ★ negative coefficients: MA5-ALG-C-01 · Gold: binomial products MA5-ALG-C-01; special products MA5-ALG-P-02

NCM 8 Adv 3.07, 3.11–3.13 → NCM 9 Adv 1.05, 1.07–1.10

KEY IDEAS

- **Distributive law:** multiply *every* term inside the brackets by the term outside: $a(b+c) = ab + ac$.
- An area model shows why: a rectangle 3 by $(x+5)$ has area $3x + 15$.
- **Negative outside:** every sign inside changes: $-2(x-5) = -2x + 10$.
- **Expand, then collect like terms:** $3(x+2) + 2(x-1) = 3x + 6 + 2x - 2 = 5x + 4$.
- **Gold –** binomial products: $(a+b)(c+d) = ac + ad + bc + bd$.

WORKED EXAMPLES

Example 1 Expand $-3(2x-4)$.

$= -3 \times 2x + (-3) \times (-4)$

$= -6x + 12$

Example 2 Expand and simplify $4(2a+1) - 3(a-2)$.

$= 8a + 4 - 3a + 6$

$= 5a + 10$

Example 3 Gold Expand $(x+3)(x+5)$.

$= x^2 + 5x + 3x + 15$

$= x^2 + 8x + 15$

Bronze round

Support · everyone starts here

Expand.

1 $3(x+4)$

2 $5(a-2)$

3 $2(3m+1)$

4 $4(2-y)$

5 $x(x+6)$

6 $a(b-3)$

7 Simplify $4x + 3 - x + 2$

↻ L1

8 $6(2p+3q)$

9 Evaluate $(-4) \times (-6)$

↻ Integers

10 $7(3-2k)$

11 $10(0.3x+2)$

12 If $a = 3$, $b = -4$: $2a + b$

↻ L1

Expand and simplify.

13 $2(x+3) + 5$

14 $4(a+1) + 3a$

15 Evaluate $\frac{2}{3} + \frac{1}{6}$

↻ Fractions

16 $y(4-y)$

17 $3(x+2) + 2(x+1)$

18 10% of 450

↻ Percentages

S Silver round

Core · aim to finish this round

Expand and simplify. ★ = beyond Stage 4 core (in your Year 8 textbook).

19 $-2(x+5)$ ★

20 $-(m+4)$ ★

21 $-4(3a-2)$ ★

22 $-x(x-7)$ ★

23 $2a(3a+5b)$

24 $5m(2-m)$

25 $3(x+2)+4(x+1)$

26 $5(2a-1)-3(a+2)$ ★

27 $x(x+3)+2(x-4)$

28 $4(3-y)-(y+2)$ ★

29 Simplify $15x^2y \div 3xy$

↻ L1

30 $-2(p-q)+3(2p+q)$ ★

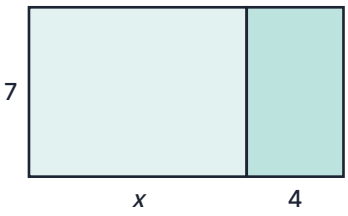
31 Share \$120 in the ratio $3:5$

↻ Ratios

32 $\frac{1}{2}(4x+6)$

33 $3x(2x-1)-x(x+4)$ ★

34 Write the area of the whole rectangle in two ways, one using brackets.



35 Jack wrote $-3(x-5) = -3x-15$. Find his error and correct it. ★

36 Adult cinema tickets cost \$$x$ and child tickets cost \$$(x-8)$.
a) Write and simplify an expression for 3 adults and 4 children.
b) Find the total if an adult ticket is \$20.

Gold round

Extension · stretch yourself

Expand and simplify.

37 $(x+2)(x+7)$

38 $(a-4)(a+6)$

39 $(2m+1)(m-3)$

40 $(x+5)^2$

41 $(y-3)(y+3)$

42 $(3a-2)^2$

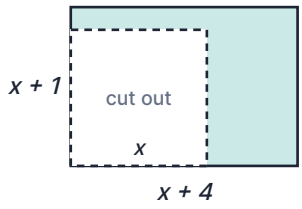
43 $(x+4)(x-4)$

44 $(2x+3)(3x-1)$

45 $(x+1)(x+2)-x(x+3)$

46 a) Show that $(n+1)^2-n^2=2n+1$.
b) Hence find 101^2-100^2 without a calculator.

47 A square of side x is cut from a rectangle measuring $(x+4)$ by $(x+1)$. Show that the area left is $5x+4$.



LESSON 3 OF 7

Factorising

Highest common factor · taking out the HCF · negative common factors · checking by expanding

50 min

No calculator

Outcomes: MA4-ALG-C-01 · MAO-WM-01 | ★ negative HCF: MA5-ALG-P-01 · Gold: binomial common factors, difference of squares, monic trinomials MA5-ALG-P-01, MA5-ALG-P-02

NCM 8 Adv 3.08–3.10 → NCM 9 Adv 1.06, 1.11–1.14

KEY IDEAS

- **Factorising** is the reverse of expanding: $6x + 15 \rightleftarrows 3(2x + 5)$.
- Find the **HCF** of the numbers *and* the pronumerals, write it outside, then divide each term by it.
- **Fully** factorised means nothing more can come out: $2x(6x - 4)$ is not finished – use $4x(3x - 2)$.
- If the first term is negative, you may take out a **negative** HCF: $-6x - 9 = -3(2x + 3)$.
- **Always check** by expanding your answer.

WORKED EXAMPLES

Example 1 Factorise $12x + 18$.

$\text{HCF} = 6$

$= 6(2x + 3)$

Example 2 Factorise $8a^2b - 12ab^2$.

$\text{HCF} = 4ab$

$= 4ab(2a - 3b)$

Example 3 Factorise $-5m - 15$.

$\text{HCF} = -5$

$= -5(m + 3)$

B Bronze round

Support · everyone starts here

Find the highest common factor (HCF).

1 12 and 18

2 $8x$ and 12

3 $6a$ and $9ab$

4 x^2 and $5x$

Factorise.

5 $3x + 12$

6 $5a - 20$

7 $4m + 6$

8 $7y - 14$

9 Expand $4(2x - 3)$

↻ L2

10 $x^2 + 3x$

11 $ab + 5a$

12 $10 - 15k$

13 Evaluate $-15 \div (-3)$

↻ Integers

14 $9p + 9$

15 Simplify $3a \times 4a$

↻ L1

16 $6xy + 8x$

17 Solve $3x + 5 = 20$

↻ Equations

18 $12a - 8b$

S Silver round

Core · aim to finish this round

Factorise fully. ★ = beyond Stage 4 core (in your Year 8 textbook).

19 $x^2 - 7x$

20 $4a^2 + 10a$

21 $9mn - 6m$

22 $-4x - 12$ ★

23 $-10a + 15$ ★

24 $-x^2 - 6x$ ★

25 $15p^2q + 20pq^2$

26 $3x^2 + 6x + 9$

27 $2a^2 - 4ab + 6a$

28 Expand $-3(2 - 5y)$ ★

↻ L2

29 A \$85 shirt is reduced by 20%. Find the sale price.

↻ Percentages

30 $18x^2y - 12xy$

31 If $x = -2$: $3x^2 - x$

↻ L1

32 Does the point $(2,\ 7)$ lie on $y = 3x + 1$?

↻ Linear

Factorise the numerator, then simplify.

33 $\dfrac{5x + 10}{5}$

34 $\dfrac{6a - 9}{3}$

35 $\dfrac{4x^2 + 8x}{4x}$

36 Sam says $12x^2 - 8x = 2x(6x - 4)$.
a) Expand to show the two sides are equal.
b) Why is Sam\'s answer not fully factorised? Correct it.

37 A rectangle has area $(6x + 15)\ \text{cm}^2$ and width 3 cm. Find an expression for its length.

38 Use factorising to evaluate $17 \times 23 + 17 \times 77$ mentally.

Gold round Extension · stretch yourself

Factorise fully.

39 $a(x+3)+b(x+3)$

40 $x(y-2)-4(y-2)$

41 x^2-25

42 $4m^2-9$

43 $x^2+7x+12$

44 $x^2-2x-15$

45 x^2+x-12

46 $9a^2-1$

47 $2x^2-18$

48 Use the difference of two squares to evaluate 53^2-47^2 without a calculator.

49 Factorise n^2+n, then explain why n^2+n is always even for any whole number n.

LESSON 4 OF 7

Indices

Index notation · prime factors · square and cube roots · index laws · the zero index

50 min

No calculator

Outcomes: MA4-IND-C-01 · MAO-WM-01 | Gold: index laws with pronumerals and the negative index MA5-IND-C-01

NCM 8 Adv 2.08, 2.10–2.12, 3.06 → NCM 9 Adv 5.01–5.05

KEY IDEAS

- In 2^5, 2 is the **base** and 5 is the **index** (power): $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.
- **Multiply:** add indices $a^m \times a^n = a^{m+n}$. **Divide:** subtract $a^m \div a^n = a^{m-n}$.
- **Power of a power:** multiply $(a^m)^n = a^{mn}$. **Products:** $(ab)^n = a^n b^n$.
- **Zero index:** $a^0 = 1$ (for $a = 0$). The laws only work when the **bases are the same**.
- **Prime factors:** use a factor tree, then write in index form: $72 = 2^3 \times 3^2$.

WORKED EXAMPLES

Example 1 Simplify, then evaluate $3^4 \times 3^5 \div 3^6$.

$= 3^{4+5-6}$

$= 3^3 = 27$

Example 2 Write 360 as a product of primes.

$360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5$

$= 2^3 \times 3^2 \times 5$

Example 3 Evaluate $(2^3)^2 \times 5^0$.

$= 2^6 \times 1$

$= 64$

B Bronze round

Support · everyone starts here

Evaluate, or write as instructed.

1 Write $5 \times 5 \times 5 \times 5$ in index form

2 2^5

3 10^4

4 $3^2 + 4^2$

5 7^0

6 Simplify $4x \times 5y$

↻ L1

7 $\sqrt{81}$

8 $\sqrt[3]{64}$

9 Factorise $8x - 20$

↻ L3

10 Which is larger: 2^5 or 5^2 ?

11 Expand $5(2a + 3)$

↻ L2

12 $\frac{3}{5} \times \frac{10}{9}$

↻ Fractions

Write as a single power.

13 $2^3 \times 2^4$

14 $5^9 \div 5^3$

15 $(3^2)^4$

16 $6^5 \div 6$

Write each number as a product of its prime factors in index form.

17 24

18 100

19 90

20 56

S Silver round

Core · aim to finish this round

Evaluate (use the index laws first).

21 $7^4 \times 7^6 \div 7^8$

22 $(2^4)^3 \div 2^{10}$

23 $\dfrac{3^5 \times 3^4}{3^7}$

24 $(2 \times 5)^3$

25 $4^0 + 4^1 + 4^2$

26 $5 \times 2^3 - (3+1)^2$

27 $2^3 \times 3^2$

28 Factorise $-6a - 9$ ★

↻ L3

29 540 as a product of primes

30 1260 as a product of primes

31 Use $1764 = 2^2 \times 3^2 \times 7^2$ to find $\sqrt{1764}$

32 If $b = -4$, evaluate $5 - 2b$

↻ L1

33 40 as a percentage of 160

↻ Percentages

34 $\sqrt{50}$ lies between which two whole numbers?

35 Find n: $2^n = 128$

36 Find n: $3^n \times 3^2 = 3^9$

37 Expand $x(x-4) + 3x$

↻ L2

38 Simplify $\dfrac{2^{10}}{2^7 \times 2^3}$

39 True or false: $2^3 \times 2^4 = 4^7$? Explain.

40 One bacterium doubles every hour.
a) How many are there after 10 hours?
b) After how many hours are there first more than $1\,000\,000$?

41 Use $5^3 \div 5^3$ in two different ways to explain why $5^0 = 1$.

Gold round

Extension · stretch yourself

Simplify. (Index laws with pronumerals.)

42 $a^4 \times a^5$

43 $x^7 \div x^2$

44 $(m^3)^4$

45 $3x^2 \times 4x^5$

46 $(2a^3)^2$

47 $\dfrac{12p^5q^2}{4p^2q}$

48 $5y^0 + (5y)^0$

49 Evaluate 2^{-3}

50 $(3x^2y)^3$

51 $\dfrac{(2a^2)^3}{4a^5}$

52 $\dfrac{x^3 \times x^4}{(x^2)^3}$

53 Evaluate 10^{-2}

54 Which is larger, 2^{30} or 3^{20}? Justify your answer without a calculator.

55 Find the value of n if $4^n = 2^{10}$.

Hint: write 4 as a power of 2.

LESSON 5 OF 7

Square roots & Pythagoras' theorem

Exact and approximate roots · hypotenuse and shorter sides · the converse · triads · surds (Gold)

50 min

No calculator

Outcomes: MA4-IND-C-01 (roots) · MA4-PYT-C-01 · MAO-WM-01 | Gold: surds MA5-IND-P-02 · length of an interval MA5-LIN-C-01 · 3D Pythagoras MA5-TRG-P-01

NCM 8 Adv 1.01–1.08, 2.09 → NCM 9 Adv 2.01–2.09

KEY IDEAS

- The **hypotenuse** is the longest side, opposite the right angle. **Pythagoras:** $c^2 = a^2 + b^2$.
- **Shorter side:** subtract: $a^2 = c^2 - b^2$.
- **Exact** answers keep the root: $\sqrt{74}$. **Estimate** a root using squares: $8^2 = 64,\ 9^2 = 81$, so $\sqrt{74}$ is between 8 and 9.
- $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$, e.g. $\sqrt{9 \times 16} = 3 \times 4 = 12$.
- **Converse:** if $c^2 = a^2 + b^2$ (with c the longest side), the triangle is right-angled. Whole-number solutions such as $3, 4, 5$ are **Pythagorean triads**.

WORKED EXAMPLES

Example 1 Find c if the shorter sides are 5 and 12.

$$c^2 = 5^2 + 12^2 = 169$$
$$c = \sqrt{169} = 13$$

Example 2 Find x if the hypotenuse is 9 and one side is 5.

$$x^2 = 9^2 - 5^2 = 56$$
$$x = \sqrt{56}\ (\text{exact})$$

Example 3 Is a triangle with sides $7, 24, 25$ right-angled?

$$7^2 + 24^2 = 49 + 576 = 625$$
$$25^2 = 625 \checkmark\ \text{Yes}$$

B Bronze round

Support · everyone starts here

Evaluate, or answer as instructed.

1 $\sqrt{49}$

2 $\sqrt{144}$

3 $(\sqrt{13})^2$

4 $\sqrt{9 \times 16}$

5 $\sqrt[3]{27}$

6 Evaluate $2^3 \times 2^2$

↻ L4

7 Simplify $7x - 2x + x$

↻ L1

8 Factorise $5y + 15$

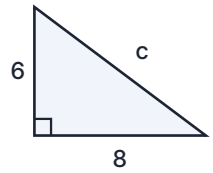
↻ L3

9 $\sqrt{30}$ lies between which two whole numbers?

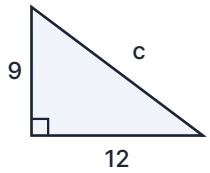
10 Estimate $\sqrt{30}$ to 1 d.p. (check by squaring)

Find the length of the hypotenuse, c. Leave answers that are not whole numbers in exact form. (Lengths in cm.)

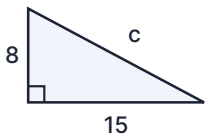
11



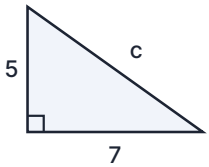
12



13



14



15 Write 0.35 as a percentage

↻ Percentages

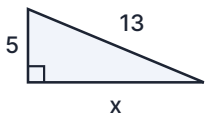
16 In $\triangle PQR$, $\angle Q = 90°$. Name the hypotenuse.

S Silver round

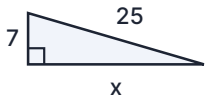
Core · aim to finish this round

Find the length of the side marked x. Leave answers that are not whole numbers in exact form. (Lengths in cm.)

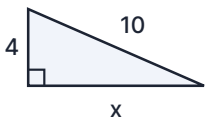
17



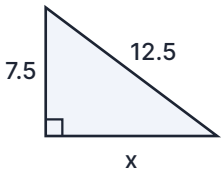
18



19



20



Answer each question.

21 Exact length of the hypotenuse if the shorter sides are 2 and 3

22 Is a triangle with sides $9, 40, 41$ right-angled?

23 Is a triangle with sides $6, 9, 11$ right-angled?

24 Is $20, 21, 29$ a Pythagorean triad?

25 Expand $-2(3x - 1)$ ★

L2

26 Evaluate $5^0 + 2^4$

L4

27 Simplify $\frac{3a}{4} \times \frac{8}{a}$

L1

28 Factorise $x^2 + 4x$

L3

29 Estimate $\sqrt{90}$ to 1 decimal place without a calculator

30 Simplify the ratio $2.5 : 1.5$

Ratios

31 A 5 m ladder leans against a wall with its foot 1.4 m from the wall. How high up the wall does it reach? *($48^2 = 2304$)*

32 A farm gate is 2.4 m wide and 1 m high. Find the length of the diagonal brace across it.

33 Ava walks 600 m east along Marine Parade, Kingscliff, then 250 m north. How much shorter would the straight-line path back to her start have been?

34 $3, 4, 5$ is a triad. Write two more triads that are multiples of it, and explain why multiples of a triad are always triads.

G Gold round

Extension · stretch yourself

Simplify. (Surds.)

35 $\sqrt{12}$

36 $\sqrt{50}$

37 $\sqrt{72}$

38 $2\sqrt{3} + 5\sqrt{3}$

39 $3\sqrt{2} \times 2\sqrt{5}$

40 $\sqrt{45} - \sqrt{20}$

41 Hypotenuse, shorter sides 4 and 8, in simplest surd form

42 $(2\sqrt{3})^2$

43 $\sqrt{2} \times \sqrt{8}$

44 $\sqrt{3}(\sqrt{3} + 1)$

45 $\dfrac{\sqrt{48}}{\sqrt{3}}$

46 $5\sqrt{2} - \sqrt{8}$

47 Find the length of the interval joining $A(1,\ 2)$ and $B(4,\ 6)$.

Hint: draw a right-angled triangle.

48 Show that a triangle with sides $\sqrt{5}$, $\sqrt{20}$ and 5 is right-angled.

49 A box measures 12 cm by 4 cm by 3 cm. Find the length of the longest straight rod that fits inside it.

LESSON 6 OF 7

Equations & formulas

Inverse operations · unknowns on both sides · $ax^2 = c$ · substituting into formulas · changing the subject (Gold)

50 min

No calculator

Outcomes: MA4-EQU-C-01 · MAO-WM-01 | Gold: 3-step equations and grouping symbols MA5-EQU-C-01 · two fractions, changing the subject MA5-EQU-P-02 *NCM 8 Adv 9.01–9.08 → NCM 9 Adv 7.01–7.09*

KEY IDEAS

- Keep the equation balanced: do the **same inverse operation** to both sides.
- **Unknowns on both sides:** first move the pronumeral terms to one side: $5x = 2x + 12 \Rightarrow 3x = 12$.
- **Check** every solution by substituting it back into the original equation.
- $x^2 = c$ (with $c > 0$) has **two** solutions: $x = \pm\sqrt{c}$, since $(-6)^2 = 6^2 = 36$.
- **Formulas:** substitute the known values first, then solve for the unknown.

WORKED EXAMPLES

Example 1 Solve $5x - 7 = 18$.

$5x = 25$

$x = 5$

Example 2 Solve $7x - 3 = 4x + 9$.

$3x - 3 = 9$

$3x = 12, \quad x = 4$

Example 3 Solve $2x^2 = 50$.

$x^2 = 25$

$x = \pm 5$

B Bronze round

Support · everyone starts here

Solve.

1 $x + 9 = 15$

2 $y - 4 = -3$

3 $6m = 42$

4 $\frac{k}{5} = 3$

5 $2x + 3 = 11$

6 $3a - 5 = 16$

7 $\frac{n}{4} + 2 = 5$

8 Expand $6(x - 2)$

↻ L2

9 $10 - x = 4$

10 $4p + 7 = -5$

11 Evaluate $3^2 \times 3^3$

↻ L4

12 $x^2 = 36$

13 Hypotenuse if the shorter sides are 3 and 4

↻ L5

14 $5x = 2x + 12$

15 Factorise $6a + 8$

↻ L3

16 $2(x + 3) = 14$

17 $\frac{x + 1}{3} = 4$

18 15% of 200

↻ Percentages

S Silver round

Core · aim to finish this round

Solve. Leave answers that are not whole numbers in exact form.

19 $6x - 5 = 2x + 11$

20 $5a + 2 = 2a - 10$

21 $\frac{3x}{4} = 9$

22 $8 - 3x = 2x - 7$

23 $6 - 2x = 14$

24 $\frac{x}{5} - 3 = -1$

25 $0.5x + 4 = 7$

26 $\frac{x - 4}{3} = 5$

27 $3x^2 = 48$

28 $x^2 + 5 = 30$

29 $5x^2 = 40$

30 Evaluate $(5^2)^3 \div 5^4$

L4

31 Expand $2(x + 4) - (x - 3)$ ★

L2

32 Is $8, 15, 17$ a Pythagorean triad?

L5

33 Write $\frac{3}{8}$ as a percentage

Percentages

34 $\frac{5}{6} - \frac{1}{4}$

Fractions

Substitute into the formula, then solve.

35 $v = u + at$. Find t when $v = 25$, $u = 5$, $a = 4$.

36 $A = \frac{1}{2}bh$. Find h when $A = 30$ and $b = 8$.

37 $E = \frac{1}{2}mv^2$. Find v ($v > 0$) when $E = 100$, $m = 8$.

38 $A = \pi r^2$. Find r ($r > 0$) when $A = 49\pi$.

39 I think of a number, multiply it by 4 and subtract 7. The answer is the same as doubling the number and adding 9. Write an equation and find the number.

40 A Tweed Heads taxi charges $\$4.50$ plus $\$2.20$ per km. A trip cost $\$37.50$. Write an equation and find the length of the trip.

Gold round Extension · stretch yourself

Solve.

41 $3(2m-1)=21$

42 $4(x+2)=3(x-1)$

43 $\dfrac{2x-3}{5}=3$

44 $\dfrac{x}{3}+\dfrac{x}{4}=7$

45 $\dfrac{x+2}{3}=\dfrac{x-1}{2}$

46 $2(3y+1)-y=27$

47 $\dfrac{3x-1}{4}=\dfrac{x+3}{2}$

48 $5(x-2)-2(x+1)=0$

49 $\dfrac{2}{x}=\dfrac{8}{12}$

Make the pronumeral in brackets the subject.

50 $y=3x+2 \quad [x]$

51 $C=2\pi r \quad [r]$

52 $v^2=u^2+2as \quad [a]$

53 The length of a rectangle is 3 cm more than twice its width. Its perimeter is 42 cm. Find its dimensions.

54 Three consecutive even numbers add to 138. Find them.

LESSON 7 OF 7

Area & volume

Rectangles, triangles, parallelograms, trapeziums, kites and rhombuses · composite areas · prisms and cylinders

50 min

No calculator

Outcomes: MA4-ARE-C-01 · MA4-VOL-C-01 · MAO-WM-01 | Gold: composite solids and algebraic area/volume (Stage 4 content, harder problems)

NCM 8 Adv 5.05–5.06, 5.09–5.13 → NCM 9 Adv 10.04–10.09

KEY IDEAS

- $A_{\text{rectangle}} = lb$ $A_{\text{triangle}} = \frac{1}{2}bh$ $A_{\text{parallelogram}} = bh$ (h is the **perpendicular** height).
- $A_{\text{trapezium}} = \frac{h}{2}(a+b)$ $A_{\text{kite/rhombus}} = \frac{1}{2}xy$ (x, y are the diagonals).
- **Composite shapes:** split into simple shapes and add, or take a piece away and subtract.
- **Prism:** $V = Ah$, where A is the area of the cross-section. **Cylinder:** $V = \pi r^2 h$.
- Leave answers in terms of π (exact), e.g. $V = 90\pi$ cm^3. Units: cm^2 for area, cm^3 for volume.

WORKED EXAMPLES

Example 1 Trapezium: $a = 6,\ b = 10,\ h = 5$ (cm).

$$A = \tfrac{5}{2}(6+10)$$
$$= 40 \text{ cm}^2$$

Example 2 Triangular prism: triangle base 6, height 4; length 10 (cm).

$$A = \tfrac{1}{2} \times 6 \times 4 = 12$$
$$V = 12 \times 10 = 120 \text{ cm}^3$$

Example 3 Cylinder: $r = 3$ cm, $h = 10$ cm.

$$V = \pi \times 3^2 \times 10$$
$$= 90\pi \text{ cm}^3$$

B Bronze round

Support · everyone starts here

Find the area or volume. Include units.

1 Rectangle 9 m by 4 m

2 Square of side 7 cm

3 Triangle: base 10 cm, height 6 cm

4 Parallelogram: base 12 cm, height 5 cm

5 Trapezium: $a = 4$, $b = 8$, $h = 5$ (cm)

6 Kite: diagonals 6 cm and 10 cm

7 Rhombus: diagonals 12 m and 7 m

8 Solve $3x + 4 = 19$

↻ L6

9 Evaluate 4^3

↻ L4

10 Rectangular prism $5 \times 4 \times 3$ (cm)

11 Cube of side 6 cm

12 Prism: cross-section 15 cm^2, height 8 cm

13 Hypotenuse: shorter sides 6 and 8

↻ L5

14 Expand $x(x+5)$

↻ L2

15 Factorise $9m - 12$

↻ L3

16 Cylinder: $r = 1$ cm, $h = 10$ cm (in terms of π)

17 Triangular prism: triangle base 5, height 4; length 9 (cm)

18 50% of 3.6

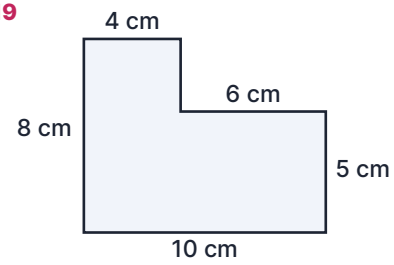
↻ Percentages

S Silver round

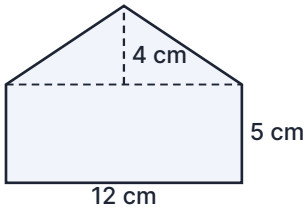
Core · aim to finish this round

Find the area of the first two shapes (drawn to scale) and the volume of the prism.

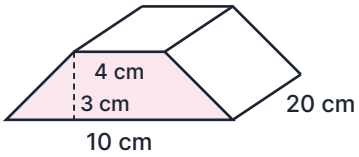
19



20



21



Volume? (not to scale)

Find the unknown. Leave volumes of cylinders in terms of π.

22 Trapezium: $a = 7.5$, $b = 12.5$, $h = 6$ (cm). Area?

23 Parallelogram: area $72\ \text{cm}^2$, base 9 cm. Height?

L6

24 Kite: area $54\ \text{cm}^2$, one diagonal 12 cm. Other diagonal?

L6

25 Cylinder: $r = 5$ cm, $h = 8$ cm. Volume?

26 Cylinder: diameter 10 cm, $h = 12$ cm. Volume?

27 Prism with the L-shape above as cross-section, depth 5 cm. Volume?

28 Isosceles triangle: base 10 cm, equal sides 13 cm. Area?

L5

29 Rectangle $3x$ by $2x$. Area?

L1

30 Rectangular prism: $V = 240\ \text{cm}^3$, base 8×6 cm. Height?

31 A pool 25 m long, 10 m wide, 1.5 m deep. Volume?

32 Rectangle 4 by $(x + 3)$. Area, expanded?

L2

33 Cube with volume $343\ \text{cm}^3$. Side length?

L4

34 Explain why the area of a triangle is half the area of a parallelogram with the same base and height.

35 A garden bed is a trapezium with parallel sides 3.5 m and 5.5 m, 2 m apart. It is filled with soil 0.3 m deep. Find the volume of soil.

Gold round

Extension · stretch yourself

36 A pipe is 100 cm long, with outer radius 5 cm and inner radius 4 cm. Find the volume of material, in terms of π.

37 A cylinder has volume 360π cm^3 and radius 6 cm. Find its height.

38 A prism has a square base of side x cm and height $(x+3)$ cm.
a) Show that $V = x^3 + 3x^2$. b) Find V when $x = 4$.

39 A trapezium has parallel sides x cm and $(x+4)$ cm, height 5 cm and area 45 cm^2. Find x.

40 A 10 cm cube has a cylindrical hole of radius 2 cm drilled straight through it. Find the volume left, in terms of π.

41 What happens to the volume of a cylinder if a) its radius is doubled? b) both its radius and height are doubled? Explain.

BRIDGE TO YEAR 9 ADVANCED

Post-test

Do this after Lesson 7. Compare your score with your pre-test.

20 marks · 20 minutes · No calculator

Name _____ Date _____ **Score** / 20

1 Algebra basics

Lesson 1 · 3 marks

a) Simplify $7m + 2n - 3m$

[1]

b) Simplify $5x \times 2xy$

[1]

c) Evaluate $3a^2 - b$ when $a = -2$ and $b = 5$

[1]

2 Expanding brackets

Lesson 2 · 3 marks

a) Expand $y(y + 5)$

[1]

b) Expand $-4(3x - 2)$

[1]

c) Expand and simplify $3(a + 4) - 2(a - 1)$

[1]

3 Factorise fully

Lesson 3 · 3 marks

a) $15k + 20$

[1]

b) $8a^2 - 12ab$

[1]

c) $-6x - 15$

[1]

4 Indices

Lesson 4 · 3 marks

a) Write $3^8 \times 3^2 \div 3^7$ as a single power, then evaluate

[1]

b) Evaluate $(5^2)^3 \div 5^4$

[1]

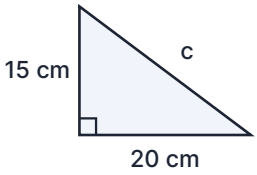
c) Write 252 as a product of prime factors in index form

[1]

5 Pythagoras' theorem

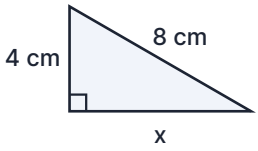
Lesson 5 · 3 marks

a) Find c.



[1]

b) Find x as an exact value.



[1]

c) Is a triangle with sides 8, 15 and 17 right-angled? Show working.

[1]

6 Solve

Lesson 6 · 3 marks

a) $3x + 8 = 29$

[1]

b) $7a - 5 = 3a + 11$

[1]

c) $3x^2 = 75$

[1]

7 Area and volume

Lesson 7 · 2 marks

a) Find the area of a trapezium with parallel sides 6 cm and 10 cm and perpendicular height 5 cm.

[1]

b) Find the volume of a cylinder with radius 2 cm and height 7 cm, in terms of π.

[1]

How much did I grow?

QUESTION	LESSON	MY MARKS	ROUNDS I COMPLETED
1 Algebra basics	Lesson 1	/ 3	Bronze ☐ Silver ☐ Gold ☐
2 Expanding brackets	Lesson 2	/ 3	Bronze ☐ Silver ☐ Gold ☐
3 Factorise fully	Lesson 3	/ 3	Bronze ☐ Silver ☐ Gold ☐
4 Indices	Lesson 4	/ 3	Bronze ☐ Silver ☐ Gold ☐
5 Pythagoras' theorem	Lesson 5	/ 3	Bronze ☐ Silver ☐ Gold ☐
6 Solve	Lesson 6	/ 3	Bronze ☐ Silver ☐ Gold ☐
7 Area and volume	Lesson 7	/ 2	Bronze ☐ Silver ☐ Gold ☐

Pre-test score: _____ / 20 → Post-test score: _____ / 20

Summary sheet

Algebra

Like terms: same pronumerals, same powers.

$4a \times 3ab = 12a^2b$, $\dfrac{12xy}{4x} = 3y$

$a(b+c) = ab + ac$, $-(x-5) = -x+5$

Factorise: take out the HCF, check by expanding.

Index laws

$a^m \times a^n = a^{m+n}$

$a^m \div a^n = a^{m-n}$

$(a^m)^n = a^{mn}$, $(ab)^n = a^n b^n$

$a^0 = 1$

Pythagoras

$c^2 = a^2 + b^2$ (c = hypotenuse)

Shorter side: $a^2 = c^2 - b^2$

Triads: $3, 4, 5$; $5, 12, 13$; $8, 15, 17$; $7, 24, 25$

$\sqrt{ab} = \sqrt{a} \times \sqrt{b}$

Equations

Do the same inverse operation to both sides.

Pronumerals on both sides: collect them on one side first.

$x^2 = c \Rightarrow x = \pm\sqrt{c}$

Always check by substituting.

Area and volume

$A = lb$ · $A = \frac{1}{2}bh$ · $A = bh$ (parallelogram) · $A = \frac{h}{2}(a+b)$ (trapezium) · $A = \frac{1}{2}xy$ (kite, rhombus)

Prism: $V = Ah$ · Cylinder: $V = \pi r^2 h$ · units: cm^2 for area, cm^3 for volume

Watch out! Common mistakes

× $3x + 2y = 5xy$ ✓ unlike terms stay apart

× $-(x-5) = -x-5$ ✓ $-x+5$

× $5^0 = 0$ ✓ $5^0 = 1$

× $\sqrt{a^2+b^2} = a+b$ ✓ $\sqrt{9+16} = 5 = 7$

× $2x(6x-4)$ is "factorised" ✓ fully: $4x(3x-2)$

× $-3^2 = 9$ ✓ $-3^2 = -9$, but $(-3)^2 = 9$

× $2^3 \times 2^4 = 4^7$ ✓ 2^7 (add indices, keep the base)

× $x^2 = 36 \Rightarrow x = 6$ ✓ $x = \pm 6$

× adding to find a shorter side ✓ $a^2 = c^2 - b^2$

× using the slant side as height ✓ use the perpendicular height

Key vocabulary

pronumeral a letter standing for a number

like terms same pronumerals, same powers

factorise write as a product, e.g. $3(x+4)$

base, index in 2^5: base 2, index 5

hypotenuse longest side, opposite the right angle

perpendicular height height at right angles to the base

coefficient the number in front of a pronumeral, e.g. 5 in $5x$

expand remove grouping symbols

HCF highest common factor

surd a root that is not a whole number, e.g. $\sqrt{7}$

triad three whole numbers with $a^2 + b^2 = c^2$

cross-section the uniform face of a prism

Coming up in Year 9 Advanced

Products and factors (binomial products, factorising quadratics) · surds · index laws with negative and fractional indices · equations and quadratic equations · coordinate geometry · trigonometry · surface area and volume. Every one of them uses the skills in this booklet.